



SEMESTER 2 EXAMINATIONS 2021/2022

MODULE: CA267 - Software Testing

PROGRAMME(S):
CASE BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY: 2

EXAMINER(S): Dr. Silvana MacMahon (Internal) (Ext:5620)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[7 Marks]**

Explain the role of exit criteria in the context of software testing and provide examples of exit criteria that are applied in software testing, clearly indicating why the criteria you highlight are important.

Q 1(b)**[7 Marks]**

Outline the concept of risk-based testing, clearly identifying different types of software risk. Provide examples that demonstrate the value of risk based testing approaches.

Q 1(c)**[11 Marks]**

Explain clearly, using examples, what is meant by Software System Decomposition. What is the reason for decomposition and what are the benefits and limitations of this approach?

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]****Q 2(a)****[9 Marks]**

Distinguish clearly between Alpha and Beta Testing using examples to illustrate your answer. In your opinion, what are the benefits and challenges associated with the two approaches? If you had to select only one of the approaches to use, which would you select and why?

Q 2(b)**[9 Marks]**

Describe clearly 3 different approaches to software integration and component integration testing. Provide an example of type of project where each of the approaches may be suitable and explain the significance of stubs and drivers in each approach.

Q 2(c)

[7 Marks]

Outline the Waterfall model for software development, and using examples, highlight the impacts for software testing.

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[9 Marks]

Research suggests that one of the benefits of Pair Programming is knowledge sharing and training. Explain using examples why this might be the case, highlighting any challenges that might arise in pair programming. Identify and discuss 2 challenges that may impact this benefit and suggest an approach to address each challenge.

Q 3(b)

[6 Marks]

In your own words, provide a description for data flow analysis and program control flow analysis, and discuss the commonalities and differences that exist in these two approaches.

Q 3(c)

[10 Marks]

You are working as a software tester and have discovered a potentially serious defect. Your manager has asked you to contact the developer to raise and discuss the defect. What factors do you need to consider before and during your interaction in order to ensure that you communicate effectively with the developer?

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[8 Marks]**

When conducting dynamic testing, both “black box” and “white box” techniques can be adopted. What are black box and white box techniques? In your opinion, for software testing, are black box or white box techniques likely to be used more commonly?

Q 4(b)**[7 Marks]**

When estimating the time and resources needed for a software testing project, would you advocate the use of contingency? Justify your response with a clear explanation and examples as appropriate.

Q 4(c)**[4 Marks]**

Through use of examples, describe how defect severity and priority can be used in combination to determine the order in which defects can be addressed.

Q 4(d)**[6 Marks]**

When should an incident be logged? Describe 3 factors that should be considered when logging an incident. Use examples to illustrate your answer.

[End of Question 4]***[END OF EXAM]***